

SOFAAMY CO. LTD

Glass & Aluminium Solutions

BUSINESS MANAGEMENT SYSTEM

System Design & Project Proposal

Prepared For:

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Sofaamy Co. Ltd, Accra, Ghana

Veloxa Technology Limited

System Design & Proposal

v1.0 — Initial Draft

1. Executive Summary

Sofaamy Co. Ltd is one of Ghana's leading glass and aluminium suppliers, providing services including glass tempering, lamination, curtain walls, shower enclosures, and aluminium fabrication to hotels, banks, hospitals, schools, and real estate developers across Accra and Kumasi.

Despite their strong market position and growing client base, Sofaamy's current operations rely entirely on manual, paper-based, and verbal communication workflows. This creates significant inefficiencies in data collection from the field, design and estimation, client communication, and internal production tracking.

Veloxa Technology Limited proposes a purpose-built Business Management System — a unified digital platform that replaces the manual workflow with a structured, real-time, role-based system covering three core areas:

#	Module	What It Solves
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1	Field & Client Intake	Reps submit structured measurements digitally from the field. Walk-in clients enter details at the office — no more lost or inaccurate data.
2	Visual Configurator & Estimator	Project supervisors generate live visual representations of doors, windows, and frameless designs with editable dimensions — and auto-produce cost estimates and engineer spec sheets.
3	Internal Job Pipeline Tracker	Every job moves through a defined chain: intake → design → payment → procurement → factory (cutting, processing, holes, assembly, glazing, QA) → delivery. Every team member sees their role in real time.

This document outlines in detail how each module works, the user roles involved, the technology stack, development phases, and recommended next steps.

2. Current Workflow & Problem Statement

Before defining the solution, it is important to understand Sofaamy's current operational flow and the specific pain points at each stage.

2.1 How Jobs Currently Flow

Sofaamy handles client jobs through two primary entry paths:

Path A — Field Measurement (Rep Dispatched)

Client Calls	Rep Dispatched to Site	Rep Takes Measurements	Rep Returns to Office	Supervisor Designs & Estimates	Client Receives Quote
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Path B — Walk-in Client with Sketch or Measurements

Client Walks In	Client Submits Sketch / Measurements	Supervisor Reviews	Corrections Made	Supervisor Designs & Estimates	Client Receives Quote
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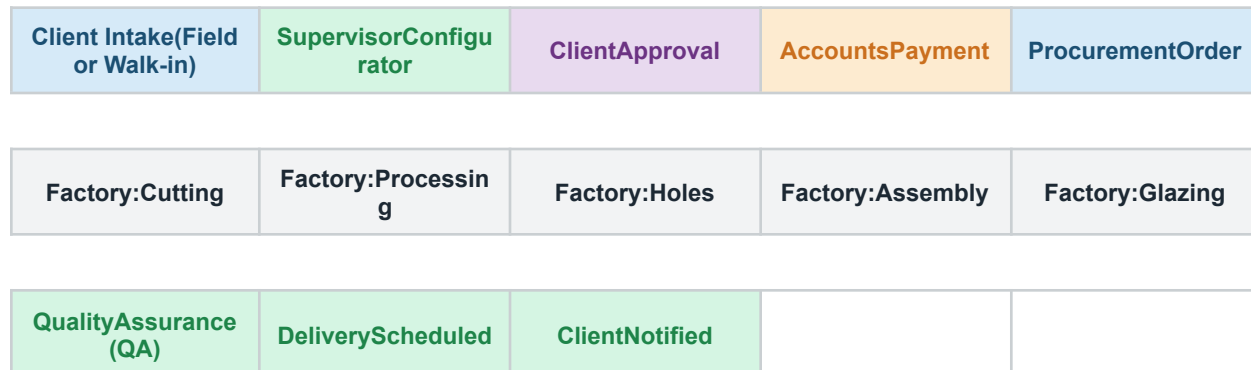
2.2 Key Problems Identified

Problem Area	Impact on Operations
Inaccurate field measurements	Reps record data on paper or verbally. Important dimensions are missed, requiring repeat site visits and delaying client quotes.
Client sketches are unreliable	Walk-in clients frequently provide wrong measurements. The supervisor must redo the design from scratch, wasting time.
Manual design & estimation	Every quote is designed manually with no visual tool. The supervisor creates the visual, the cost estimate, and the engineer spec separately — tripling the work.
No real-time job visibility	There is no central system to track where a job is. Accounts, procurement, and factory all operate in silos.
No client communication trail	Client updates are done informally over calls or WhatsApp with no record. Missed updates affect trust and satisfaction.
Factory pipeline is untracked	There is no visibility into which stage of production a job is at. Cutting, processing, glazing, and QA are managed manually.

3. Proposed Solution — System Overview

The proposed system is a web-based platform accessible on desktop, tablet, and mobile. It is divided into three integrated modules, all connected to a single database, with role-based access so each user sees only what is relevant to their job.

Overall System Flow



Module 1 — Field & Client Intake

This module replaces paper-based and verbal data collection with a structured digital form accessible from any phone, tablet, or computer.

How It Works — Field Rep (On-Site):

- The project supervisor dispatches a rep to the client's project site as usual.
- The rep opens the Sofaamy app on their phone — no installation required, it works in the browser.
- The rep fills a structured intake form: client name, site address, type of product needed (window, door, frameless, etc.), dimensions for each opening, number of units, and on-site photos.
- Once submitted, the data is instantly visible to the supervisor in the office — no waiting for the rep to return.
- The supervisor receives a notification immediately.

How It Works — Walk-In Client:

- The supervisor or front desk staff opens a new job and enters the client's measurements directly into the system.
- If the client's measurements appear inconsistent with standard product sizes, the system flags it automatically.
- The supervisor can then adjust and correct the measurements before proceeding.

What the Intake Form Captures:

Field	Details
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Client Information	Name, phone number, location, company (if applicable)
Product Type	Window frame, door, frameless glass, curtain wall, etc.
Dimensions	Width, height, depth per opening — each unit measured separately
Quantity	Number of units of each product type
Material Preference	Aluminium profile type, glass type (clear, frosted, tinted, tempered)
Site Photos	Rep uploads photos directly from the phone camera
Notes	Any special instructions or observations from the site

Module 2 — Visual Configurator & Estimator

This is the centrepiece of the platform. The Visual Configurator allows the project supervisor to generate a live, accurate, editable visual representation of any product, with automatic cost estimation and engineer documentation — all from a single screen.

Step-by-Step Supervisor Workflow:

1. Open Job	2. Select Template	3. Enter Dimensions	4. Edit Live Visual	5. Generate Output
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Step 1 — Open Job from Intake:

The supervisor opens the submitted job from the intake module. All the client's measurements and product details are already populated — no re-entry needed.

Step 2 — Select a Product Template:

The system provides a library of standard Sofaamy product templates including:

- Window Frames — single pane, double pane, casement, sliding, louvre
- Door Frames — single leaf, double leaf, sliding, pivot
- Frameless Glass — shower enclosures, balustrades, partitions
- Curtain Walls — standard grid, unitised
- Custom — supervisor can start from a blank canvas

Each template comes pre-loaded with standard industry dimensions and material defaults, which the supervisor then customises.

Step 3 — Enter & Edit Dimensions:

The supervisor inputs or adjusts the width, height, frame thickness, panel count, and opening direction. Every change is reflected LIVE in the visual on the right side of the screen — no need to redraw or export. The supervisor keeps adjusting until the visual matches the client's requirement exactly.

Step 4 — Customise Design:

Once dimensions are confirmed, the supervisor can:

- Select glass type — clear, frosted, tinted, tempered, laminated
- Select aluminium finish — mill finish, anodised, powder-coated with colour options
- Add hardware — handle positions, hinge count, locking mechanism type
- Adjust panel layout — drag dividers to reposition

Step 5 — Auto-Generate Three Outputs at Once:

Output	Description
Client Visual + Quote PDF	A professional PDF showing the visual design, labelled dimensions, material selections, unit cost, and total price estimate. Ready to send to the client via WhatsApp or email.
Engineer Spec Sheet	A technical document with full dimensions, cut sizes, material quantities, glass thickness, and hardware counts. This goes directly to the factory floor.

Bill of Materials (BOM)	An itemised list of every material needed: aluminium profile lengths, glass pane sizes, screws, hinges, sealant — with quantities and estimated costs. Goes to procurement.
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The key benefit: the supervisor does one piece of work — the design — and the system produces three separate documents automatically. Currently, these three documents are each produced manually, tripling the supervisor's workload.

Module 3 — Internal Job Pipeline Tracker

Once a design is approved and payment is confirmed, the job enters the internal production pipeline. This module gives every team member — accounts, procurement, factory floor, QA, and delivery — a real-time view of every job and their specific responsibilities.

The Complete Job Pipeline:

Pre-Production:

Job Created	Design Approved	Quote Sent to Client	Client Approves	Accounts: Payment Confirmed
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Factory Production:

ProcurementOrdered	Factory:Cutting	Factory:Processing	Factory:Holes	Factory:Assembly
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Finishing & Delivery:

Factory:Glazing	QualityAssurance (QA)	DeliveryScheduled	Delivered toClient	JobClosed
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How the Pipeline Works:

- Each stage is a column on a visual job board (like a Kanban board). Cards represent individual jobs.
- When a team member completes their stage, they move the job card to the next stage with one click.
- The next person in the chain receives an instant notification on their device.
- No job can skip a stage — Accounts must confirm payment before the job is released to procurement. Procurement must submit before factory receives the order.
- Every stage is timestamped — management can see how long each stage takes and identify bottlenecks.
- The supervisor and management can view ALL jobs across all stages at any time from the dashboard.

Role-Based Visibility:

User Role	What They See
Field Rep	Mobile form for job intake. Can view status of jobs they submitted.
Project Supervisor	Full configurator. All jobs assigned to them. Client communication tools.
Accounts Team	Jobs awaiting payment confirmation. Payment status dashboard.
Procurement	Jobs released for material ordering. Bill of Materials per job.
Factory Floor	Jobs in their specific stage. Engineer spec sheet per job. No financial data.

QA Team	Jobs reaching QA stage. Checklist for each product. Approve or flag for rework.
Management	Full dashboard — all jobs, all stages, financial summary, team performance, turnaround times.

4. Client Communication

Currently, client updates are done informally over phone calls and WhatsApp with no record. The proposed system integrates structured client communication directly into the workflow.

- When a quote is ready, the supervisor sends it directly from the system via WhatsApp or email — with the visual and cost breakdown attached.
- The client receives a professional PDF with their name, the product visual, dimensions, and the price.
- If the client wants adjustments, the supervisor updates the configurator and sends a revised quote — all tracked in the system.
- When the job is delivered, the client receives an automatic delivery notification.
- All client communication is logged against the job — so there is always a record of what was agreed.

Client Communication Flow:

Supervisor Completes Design	System Generates Quote PDF	Sent via WhatsApp / Email	Client Reviews & Approves	Record Saved in System
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6. Development Phases

To manage scope, risk, and cost effectively, the system will be delivered in four phases. Each phase delivers a fully functional component that Sofaamy can begin using immediately — without waiting for the entire system to be complete.

Phase	Module	Deliverables	Est. Duration
1	Field Intake + Basic Configurator (Window Frame Pilot)	Rep mobile form, walk-in intake, basic window frame configurator with live dimension editing and PDF output	4–5 weeks
2	Full Template Library + Cost Estimation Engine	All product types (doors, frameless, curtain walls), colour/finish options, BOM and cost auto-calculation, client PDF with Sofaamy branding	4–5 weeks
3	Internal Pipeline Tracker	Full job board with all pipeline stages, role-based dashboards for all team members, stage notifications, timestamp tracking	3–4 weeks
4	Client Communication + Reporting	WhatsApp/email quote delivery, client approval tracking, management dashboard with analytics, job history, performance reports	3–4 weeks

Each phase will include a review session with the Sofaamy team before proceeding to the next. This ensures the system is built exactly to their requirements at every step.

7. Next Steps

To move forward, the following steps are recommended:

#	Action	Owner
1	Sofaamy team reviews this document and confirms the system scope aligns with their expectations	Sofaamy Team
2	Discovery session: Veloxa maps all product templates, pricing logic, and team user roles in detail	Veloxa + Sofaamy Supervisor
3	Sofaamy provides sample product specifications and current manual pricing guide	Sofaamy Team
4	Veloxa delivers formal proposal with pricing and timeline based on confirmed scope	Veloxa Technology
5	Agreement signed. Phase 1 development begins with window frame configurator as the pilot	Both Parties

8. About Veloxa Technology

Veloxa Technology Limited is a software development company based in Accra, Ghana, specialising in Softwares, AI-powered and automation-first business solutions for African enterprises. We build custom digital tools that solve real operational problems, not generic software adapted from foreign markets.

We understand the Ghanaian business environment, the infrastructure constraints, and the specific needs of companies operating in manufacturing, construction supply, and field services. Our work is built to work in Ghana. reliably, affordably, and at scale.

Veloxa Technology Limited — Building Africa's Digital Infrastructure

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